

23.04.2026

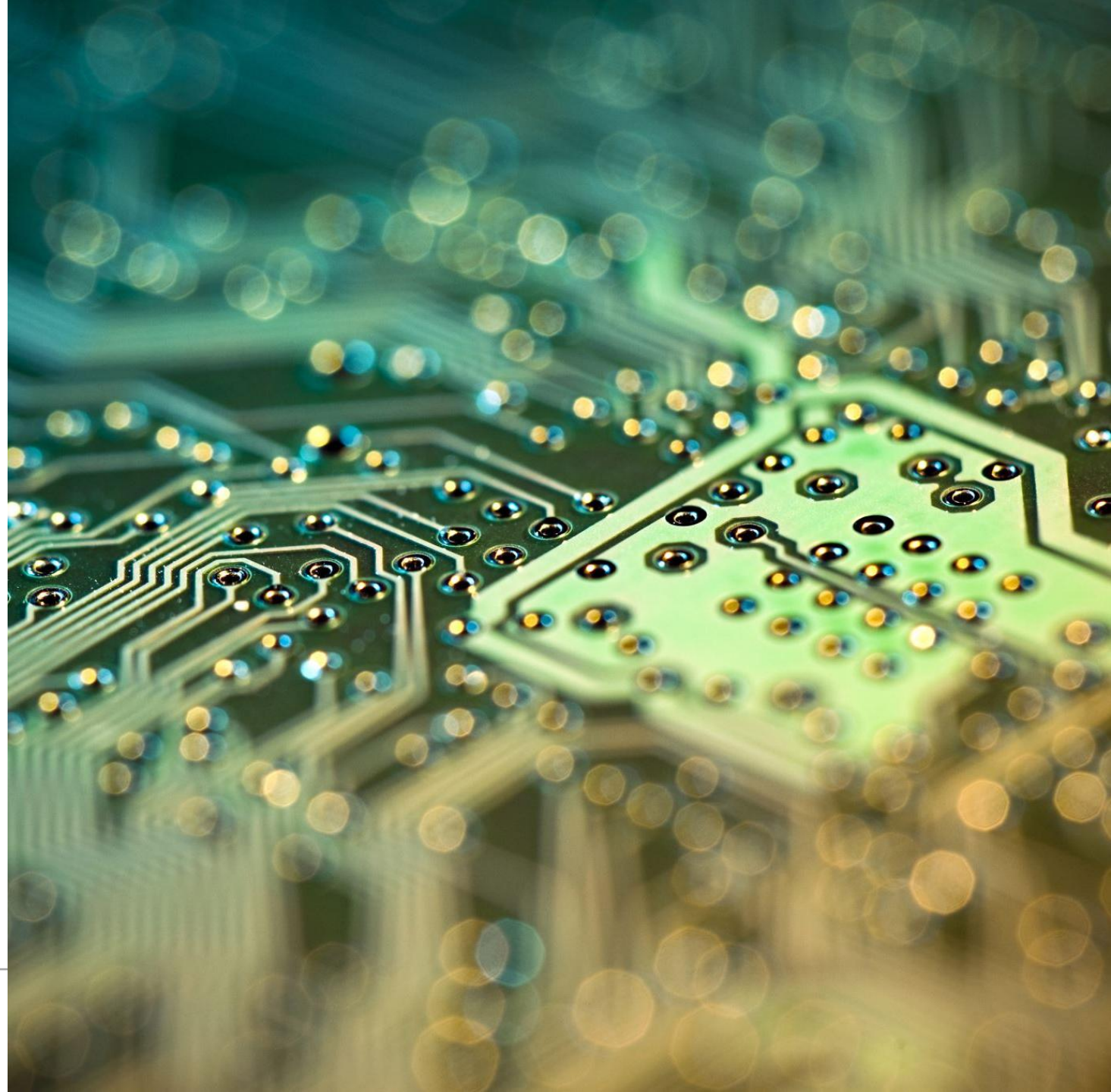
Power Electronics

(EEE/INSTR F342)

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Department of EEE

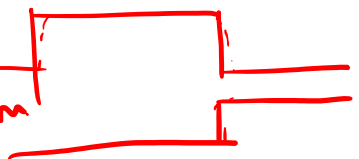
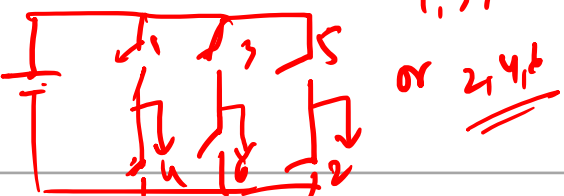


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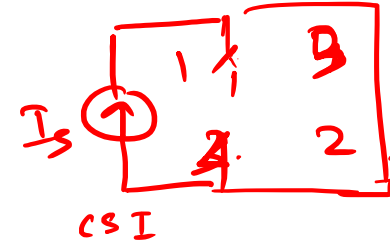
Current Source Inverters

Voltage Source Inverters (VSI)

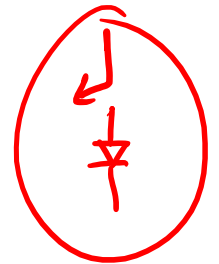
- 1) Input $\rightarrow$ voltage source
 - 2) Voltage i/p cannot be reversed $\rightarrow$ SC.
 - 3) Current i/p can be reversed $\rightarrow$
 - 4) We cannot turn ON 2 switches of the same leg together.
 - 5) Top  $\Rightarrow$ Dead Band
- 3 ϕ VSI.
- 
- 1, 3, 5
or 2, 4, 6

Current Source Inverters (CSI)

- 1) Input current source.
- 2) Current i/p cannot be reversed
- 3) But i/p. voltage can be "

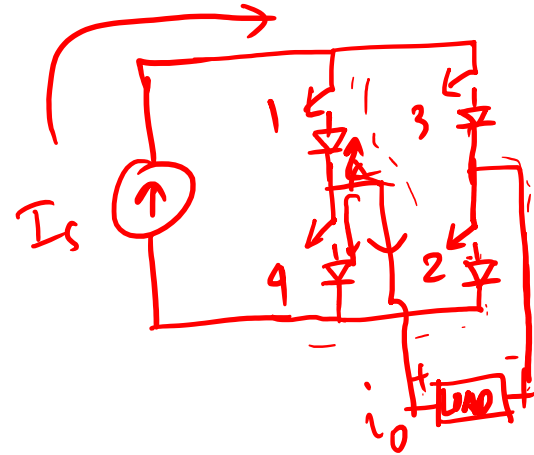
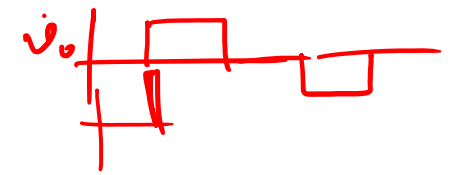


- 4) At least one of the top & one of the bottom switch should be ON. $\rightarrow$ otherwise $OC \rightarrow V_{in} = \infty$
- 5) overlap. $\rightarrow$ has to be there

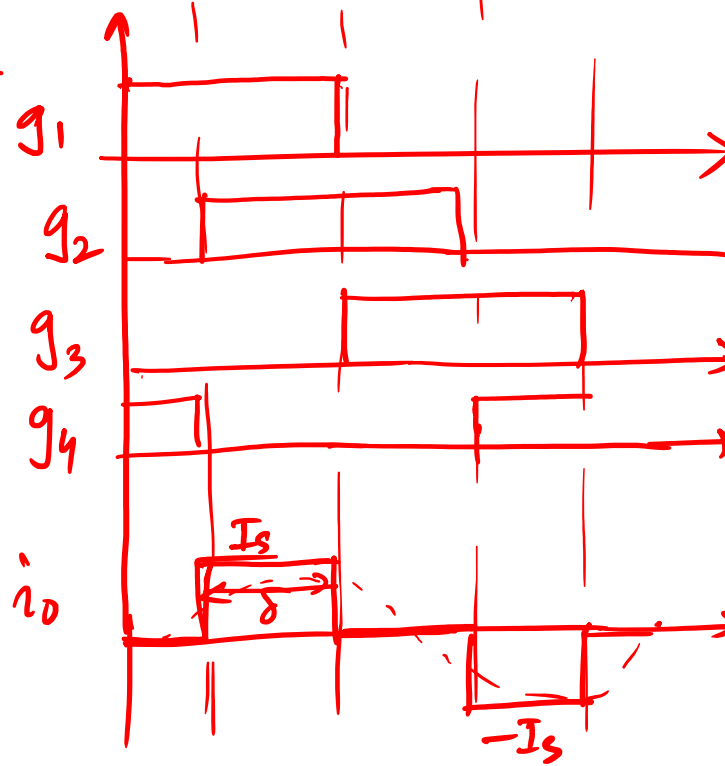


Current Source Inverters

$\frac{1}{s}$



series diode

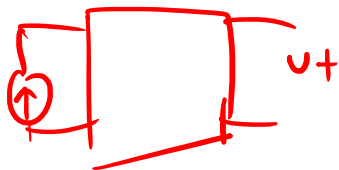
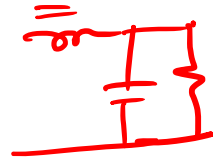
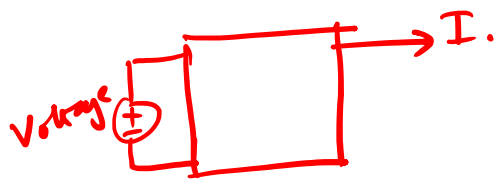


$$i_o = \sum_{n=1,3,5}^{\infty} \frac{4I_s}{n\pi} \sin \frac{n\delta}{2} \sin n\omega t$$

$$\gamma = \frac{1}{Z} = j\omega C + \frac{1}{R}$$

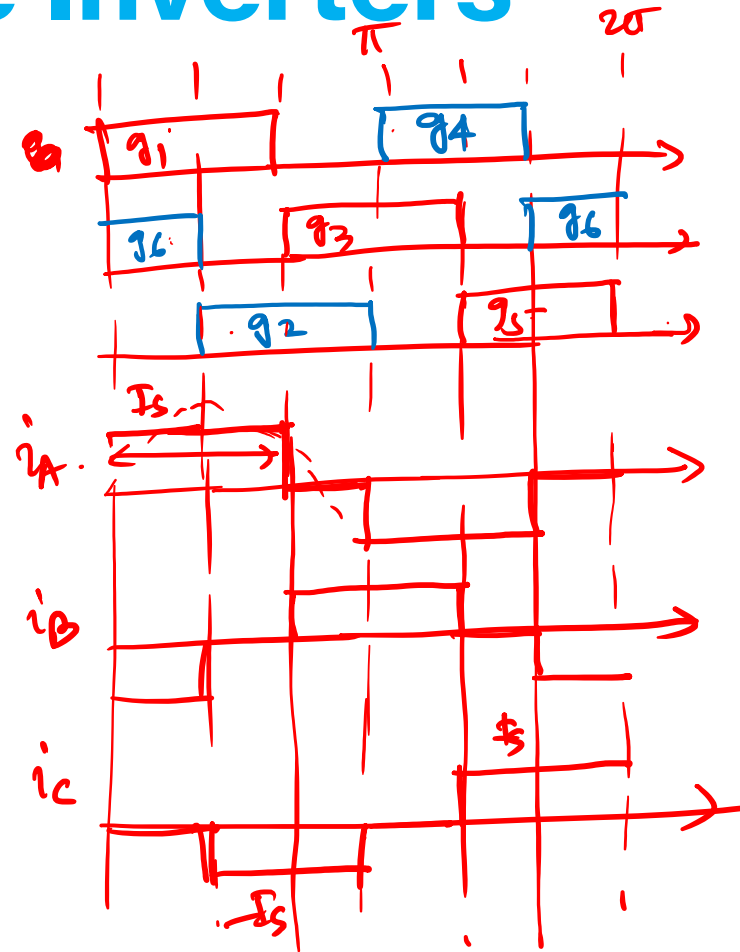
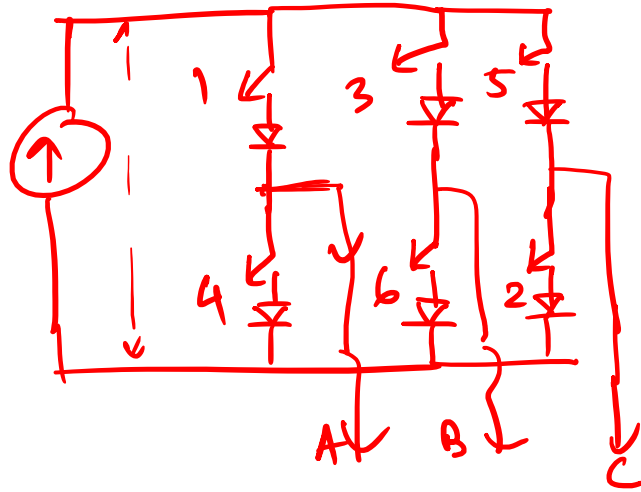
$$v_o = \sum z_n i_n$$

1φ



Current Source Inverters

3 ϕ CSI



$$i_A = \sum_{n=1,3,5}^{\infty} \frac{4I_s}{n\pi} \sin \frac{n\pi}{3} \sin n\omega t$$

$$\delta = \frac{2\pi}{3}$$

Current Source Inverters

